D02 · Capacity & Long-Range Planning

# 02 · Long-range capacity planning (12–36 months)

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## 1 WHAT IT IS

Long-range capacity planning is the structured projection of available capacity against expected demand across a 12–36 month horizon — far enough out that capex, qualifications, hiring, and outsourcing decisions can still be made, but close enough that the demand assumptions are not science fiction. It is the bridge between the strategic plan (3–5 years, often disconnected from operations) and the tactical S&OP (12–18 months, often too late to react).

## 2 WHY IT MATTERS

It is the only horizon where capacity decisions are still cheap. **Every capacity investment has a lead time** — qualifying a new production line takes 9–18 months in regulated industries, a new tester in semiconductor backend takes 26–52 weeks, hiring and training a skilled operator pool takes 6–12 months. By the time the S&OP signals a gap inside its 12–18 month window, the option set has already shrunk to overtime, co-manufacturing and customer allocation — none of which add structural capacity. The literature on operations strategy treats this as a generic principle **(Hayes & Wheelwright, 1984 — foundational text)**: **capacity is committed in years and consumed in weeks**, and the asymmetry only resolves through disciplined long-range planning.

## 3 HOW IT IS DONE

1. **Project demand by family across the 12–36 month horizon — with explicit scenarios.** Use the demand consensus as the base case; build optimistic (+15–25%) and conservative (-15–25%) scenarios anchored in commercial pipeline, design wins, market signals, not arbitrary deltas.
2. **Inventory the installed capacity that will actually exist month by month.** Account for planned shutdowns, qualification timelines of equipment already in motion, attrition in skilled labor, and contractual capacity expiration with co-manufacturers. Static capacity inventories are the most common silent failure.
3. **Compute the gap by critical resource, by scenario, by month.** Capacity - demand, expressed as utilization percentage and absolute units. A 92% utilization at month 18 in the base case but 118% in the optimistic case is a decision point, not a planning detail.
4. **Classify each gap by lead time required to close it.** Overtime closes in days; second-shift in weeks; co-manufacturer qualification in months; new line capex in 12–24 months. The gap must be visible before its closure window closes.
5. **Generate option portfolios for each gap, not single recommendations.** A single recommendation forces a binary choice; an option portfolio (cheap-and-flexible vs. expensive-and-permanent vs. hybrid) preserves decision-maker agency. This is also where the capex case (cap_03) gets triggered.
6. **Refresh the long-range view quarterly, not annually.** Annual reviews miss the inflection points that matter — a new customer design win in Q2 may already have closed the capex window by Q4. Quarterly cadence keeps the option set open.

### IN PRACTICE

**Run the 12–36 month view quarterly with scenarios**

Refresh the long-range capacity view every quarter against the scenario set from S&OP, not once a year in the budget. A rolling, scenario-based long-range plan keeps the capacity decision continuously in view; an annual one guarantees the gap is found between refreshes, when it is too late.

**Classify and time-phase every gap**

For each gap, record the lead time of its cheapest structural remedy and back-schedule the decision date. This converts the long-range plan into a calendar of decisions, each with a deadline set by the lead time of its remedy — the discipline that prevents a long-lead gap from being noticed too late.

### Feed the capex calendar

The long-range capacity analysis is the upstream input to the capital plan. When capex is driven by this analysis it is an instrument of strategy; when it is not, capex becomes a reaction to a crisis the long-range plan should have surfaced quarters earlier. The two processes must be explicitly linked.

### Connect to footprint and nearshoring context

Long-range capacity decisions interact with where capacity is placed — including nearshoring and regionalization shifts. Keep the capacity-timing analysis connected to the footprint conversation, while leaving the detailed network design and resilience modeling to its own domain. The two inform each other but are not the same decision.

#### WHAT YOU WOULD USE

**LONG-RANGE CAPACITY / STRATEGIC PLANNING**

**o9 / SAP IBP / Kinaxis — enterprise tier-1** — Long-horizon capacity modeling with scenario simulation, integrated with the S&OP/IBP plan and the financial bridge. **(o9, SAP, Kinaxis — vendor)** The choice when the long-range view must reconcile with both the operational plan and the capex calendar.

**Anaplan / Board — strategic capacity & connected planning** — Flexible connected-planning platforms strong at long-range scenario and capacity-investment modeling. **(Anaplan, Board — vendor)** A strong fit where the long-range capacity question is tightly coupled to financial planning.

**Strategic-footprint / capacity-modeling specialists** — Dedicated long-range capacity and footprint-analysis tools for complex, multi-site strategic decisions. **(specialized vendors — vendor)** The vertical option when the analysis spans many sites and a long horizon beyond the integrated plan's reach.

**For operations that don't need a platform** — A rolling 12–36 month capacity model in a workbook with explicit demand scenarios and a decision calendar back-scheduled from remedy lead times. The discipline of timing the decision to the lead time — not the tool — is what keeps the gap cheap to close.

## 4 SO WHAT

**The capacity gap that surfaces inside the S&OP horizon is a gap whose closure window has already closed.** Run the 12–36 month view quarterly with explicit scenarios, classify gaps by the lead time required to close them, and generate option portfolios — not single recommendations. The capex calendar follows this analysis; without it, capex becomes a reaction to crisis rather than an instrument of strategy.

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